

PID Library

1.0.1

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Chapter 1

Namespace Index

1.1 Package List

Here are the packages with brief descriptions (if available):

PIDLibrary		5
PIDLibrary.Control		5

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

PIDLibrary.Control.PIDController	
The base type for any PID controllers.	7
PIDLibrary.Control.PIDRotationController	
Provides functionality for computing the required acceleration to orient a Rigidbody.	9

Chapter 3

Namespace Documentation

3.1 PIDLibrary Namespace Reference

3.2 PIDLibrary.Control Namespace Reference

Classes

- class [PIDController](#)
The base type for any PID controllers.
- class [PIDRotationController](#)
Provides functionality for computing the required acceleration to orient a Rigidbody.

Chapter 4

Class Documentation

4.1 PIDLibrary.Control.PIDController Class Reference

The base type for any PID controllers.

Public Member Functions

- **PIDController** (float pKp, float pKi, float pKd)
- float **Update** (float pError, float pDeltaTime)
Updates the PID controller and returns the output.
- void **SetIntegral** (float pIntegral)
Sets the underlying 'integral' component of the PID controller.
- void **SetDerivative** (float pDerivative)
Sets the underlying 'derivative' component of the PID controller.
- void **SetPreviousError** (float pPreviousError)
Sets the underlying 'previous error' component of the PID controller.

Public Attributes

- float **Kp**
The proportion tuning value.
- float **Ki**
The integral tuning value.
- float **Kd**
The derivative tuning value.

Properties

- float **Integral** [get, set]
The integral for the PID control loop.
- float **Derivative** [get, set]
The derivative for the PID control loop.
- float **PreviousError** [get, set]
The previous error for the PID control loop.

4.1.1 Detailed Description

The base type for any PID controllers.

Author: Mathew Aloisio

4.1.2 Member Function Documentation

4.1.2.1 SetDerivative()

```
void PIDLibrary.Control.PIDController.SetDerivative (
    float pDerivative )
```

Sets the underlying 'derivative' component of the PID controller.

Parameters

<i>pDerivative</i>	
--------------------	--

4.1.2.2 SetIntegral()

```
void PIDLibrary.Control.PIDController.SetIntegral (
    float pIntegral )
```

Sets the underlying 'integral' component of the PID controller.

Parameters

<i>pIntegral</i>	
------------------	--

4.1.2.3 SetPreviousError()

```
void PIDLibrary.Control.PIDController.SetPreviousError (
    float pPreviousError )
```

Sets the underlying 'previous error' component of the PID controller.

Parameters

<i>pPreviousError</i>	
-----------------------	--

4.1.2.4 Update()

```
float PIDLibrary.Control.PIDController.Update (
    float pError,
    float pDeltaTime )
```

Updates the PID controller and returns the output.

Parameters

<i>pError</i>	
<i>pDeltaTime</i>	

Returns

the output of the updated PID controller.

The documentation for this class was generated from the following file:

- PIDController.cs

4.2 PIDLibrary.Control.PIDRotationController Class Reference

Provides functionality for computing the required acceleration to orient a Rigidbody.

Public Member Functions

- **PIDRotationController** (float pKp, float pKi, float pKd)
- void **SyncPIDs** ()
Syncs all PID tuning parameters.
- void **SyncPIDsIfDirty** ()
Syncs all PIDs only if the tuning values are out of sync.
- [PIDController](#) **GetPIDController** (int pIndex)
Returns a reference to the underlying PID controller in the given index. No bounds checking is performed.
- Vector3 [ComputeRequiredAngularAcceleration](#) (Quaternion pFromRotation, Quaternion pToRotation, Vector3 pCurrentAngularVelocity, float pDeltaTime)
Computes the required angular acceleration to rotate the object associated with the PID loops from the rotation, pFromRotation, to the target rotation, pToRotation.
- Quaternion [Update](#) (Quaternion pError, Quaternion pDelta, float pDeltaTime)
Updates the PID controllers and returns the result.

Public Attributes

- float **Kp**
The proportion tuning value.
- float **Ki**
The integral tuning value.
- float **Kd**
The derivative tuning value.

Properties

- int **PIDCount** [get]
Returns the number of PID control loops that make up the rotation controller. Use [GetPIDController\(int pIndex\)](#) to get a reference of underlying PID controllers within this range. The PID loop index order is: 0 - x, 1 - y, 2 - z, 3 - w

4.2.1 Detailed Description

Provides functionality for computing the required acceleration to orient a Rigidbody.

Author: Mathew Aloisio

4.2.2 Member Function Documentation

4.2.2.1 ComputeRequiredAngularAcceleration()

```
Vector3 PIDLibrary.Control.PIDRotationController.ComputeRequiredAngularAcceleration (
    Quaternion pFromRotation,
    Quaternion pToRotation,
    Vector3 pCurrentAngularVelocity,
    float pDeltaTime )
```

Computes the required angular acceleration to rotate the object associated with the PID loops from the rotation, pFromRotation, to the target rotation, pToRotation.

Updates the underlying PIDs.

Parameters

<i>pFromRotation</i>	
<i>pToRotation</i>	
<i>pCurrentAngularVelocity</i>	
<i>pDeltaTime</i>	

Returns

the angular acceleration required to rotate the relevant object from the specified rotation to the desired rotation under the current angular velocity condition and relevant time delta between PID updates.

4.2.2.2 GetPIDController()

```
PIDController PIDLibrary.Control.PIDRotationController.GetPIDController (
    int pIndex )
```

Returns a reference to the underlying PID controller in the given index. No bounds checking is performed.

Parameters

<i>pIndex</i>	
---------------	--

Returns

the underlying PID controller in the given index.

4.2.2.3 Update()

```
Quaternion PIDLibrary.Control.PIDRotationController.Update (
    Quaternion pError,
    Quaternion pDelta,
    float pDeltaTime )
```

Updates the PID controllers and returns the result.

Parameters

<i>pError</i>	
<i>pDelta</i>	
<i>pDeltaTime</i>	

Returns

the result of the latest PID update.

The documentation for this class was generated from the following file:

- PIDRotationController.cs

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